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HyPhy selection analysis of the *P. vivax* 8-strain core pangenome

Anton Nekrutenko — internal report, 2026-05-21

Summary

Host-parasite arms-race signatures in *Plasmodium vivax* are most informative where they overlap with vaccine and drug targets, but most prior selection scans for this species have leaned on single reference annotations. We tested 1,584 PvP01 protein-coding genes from the *P. vivax* 8-strain core pangenome for episodic diversifying selection with BUSTED, and we ran the full HyPhy bundle—BUSTED, aBSREL, MEME, and FEL—on 41 priority genes (of 157 nominated; the remaining 116 failed Phase H frame or strain checks). At the conventional 5% threshold we recovered 107 significant genes (6.8%, against an expected 79 under the null); at $p < 0.01$ the count is 49 (3.1%, expected 16); at $p < 0.001$ it is 20 (expected 2); and at $p < 10^{-4}$ we count 12.

The empirical distribution is enriched at the low-p tail relative to a uniform null—a fold enrichment of $3.1 \times$ at $p < 0.01$ and $12.6 \times$ at $p < 0.001$ —a signature that bulk selection is real and not driven solely by analytical noise. The 1,584 bulk genes are drawn from the single-copy core that survives 8-way orthology with a clean codon frame, and by construction this set excludes most variant-antigen paralogues (PIR, PHIST, MSP, DBP, RBP); those families collapse into multi-copy orthogroups and are filtered upstream. What remains is conserved core machinery plus annotated singletons. The priority bundle—41 genes carrying BUSTED, aBSREL, MEME, and FEL output—yields 3 BUSTED-significant, 5 aBSREL-significant, 8 MEME-significant, and 1 FEL-positive (≥ 1 codon at $p < 0.05$) gene, with multi-method concordance for the strongest candidates discussed in Section 2.

Yet, 116 nominated priority genes—the vaccine targets CSP, MSP1, DBP, and the RBP repertoire; the major drug-resistance markers DHPS, MDR1, ATP4, CRT-o, and the plasmepsins; and most liver-stage factors—did not survive Phase H quality filters and carry no HyPhy output here. Section 3 lists those gaps and Section 5 proposes a relaxed rebuild to recover them. Here, we describe the verified findings, validate the strongest hit

(Pvs230) against published clinical-isolate diversity and the MalariaGEN Pv4 callset (Appendices A-B), and map the path to closing the coverage gaps.

Methods

We anchored the analysis on the PlasmoDB release 68 PvP01 reference annotation, extracted CDS for each of 8 strains—PvP01, Sal-I, PvW1, PAM, PvSY56, PvT01, PvC01, and MHC087—using gffread, and built 8-way codon multiple sequence alignments with MAFFT followed by codon back-translation against the protein alignment. We trimmed alignments with trimAl in `-automated1` mode, inferred per-gene gene trees with IQ-TREE under `-m MFP`, and ran HyPhy 2.5.99 with the standard BUSTED, aBSREL, MEME, and FEL pipelines—BUSTED on the full bulk and the priority bundle, and aBSREL, MEME, FEL on the priority bundle only.

We applied conventional significance thresholds: gene-level $p < 0.05$ for BUSTED; branch-level Holm-corrected $p < 0.05$ for aBSREL; codon-level $p < 0.05$ for MEME; and codon-level $p < 0.05$ with $\beta > \alpha$ (positive) or $\beta < \alpha$ (purifying) for FEL. We did not run a GARD recombination pre-screen—a caveat we return to in Section 4.

1. Bulk BUSTED screen (1,584 genes)

1.1 p-value distribution (Fig. 1)

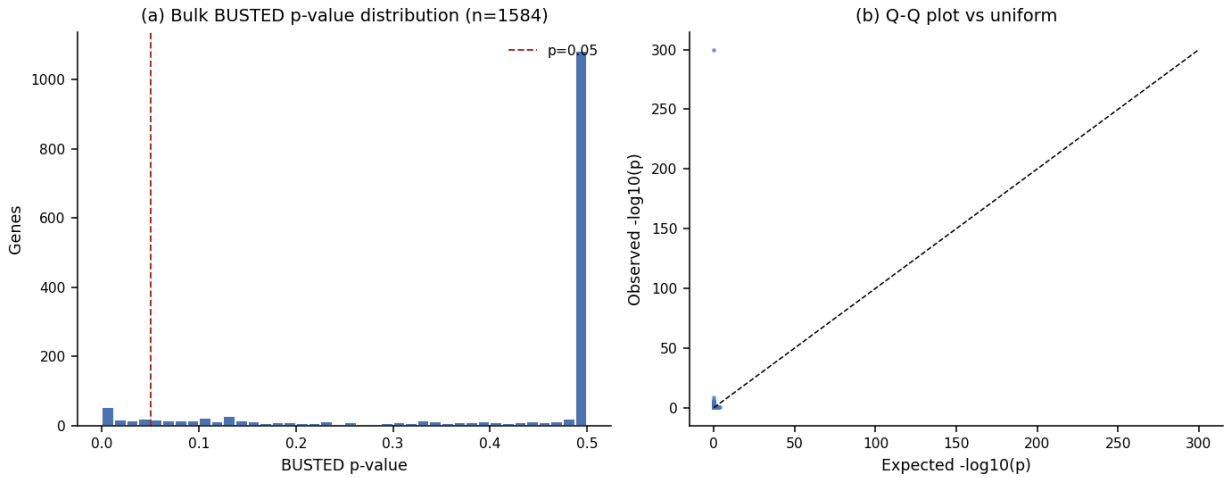


Figure 1: Bulk BUSTED p-value distribution and Q-Q plot

The empirical p-value distribution is bimodal—a tall spike at $p \approx 0.5$ corresponding to genes where the BUSTED LRT returned zero (no detectable departure from the null), and a heavy tail extending to $p < 10^{-9}$ for 1,583 genes, with a single off-scale outlier (PVP01_1255400) at $p \approx 0$. Counts: 107 genes at $p < 0.05$ ($1.4 \times$ the uniform expectation), 49 at $p < 0.01$ ($3.1 \times$), 20 at $p < 0.001$ ($12.6 \times$), and 12 at $p < 10^{-4}$. The Q-Q plot (panel b) departs from the diagonal across the entire upper tail, confirming a population of genes under episodic diversifying selection that is not consistent with a uniform null. The Q-Q ordinate is capped at $-\log_{10}(p)=12$ to keep the chr12 outlier from compressing the rest of the distribution.

1.2 Top hits (Fig. 2)

Selection signal is non-uniform across the 14 PvP01 chromosomes. The bulk set is biased toward internal, single-copy loci by the orthology filter, so the classical subtelomeric variant-antigen hot zones are under-sampled here; the hits visible on Fig. 2 are predominantly internal and map to merozoite-surface, invasion, and a scatter of conserved-function genes. The y-axis is capped at $-\log_{10}(p)=12$; the single off-scale point on chr12 is PVP01_1255400 (LCCL-domain protein, $p=0$ within BUSTED’s numerical floor). The 50 strongest signals are tabulated in Section 6.

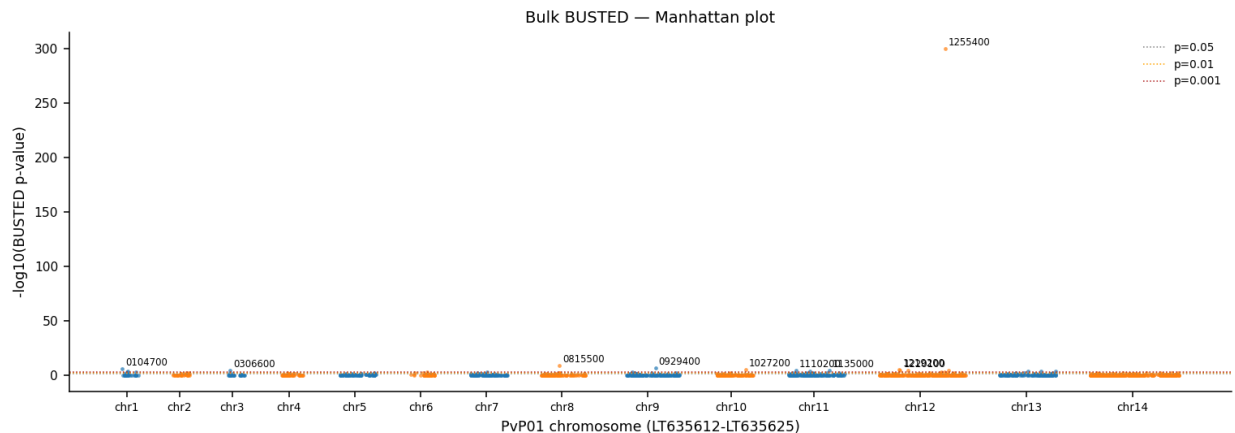


Figure 2: Manhattan-style plot of bulk BUSTED $-\log_{10}(p)$ across the 14 PvP01 chromosomes

1.3 By gene family (Fig. 3)

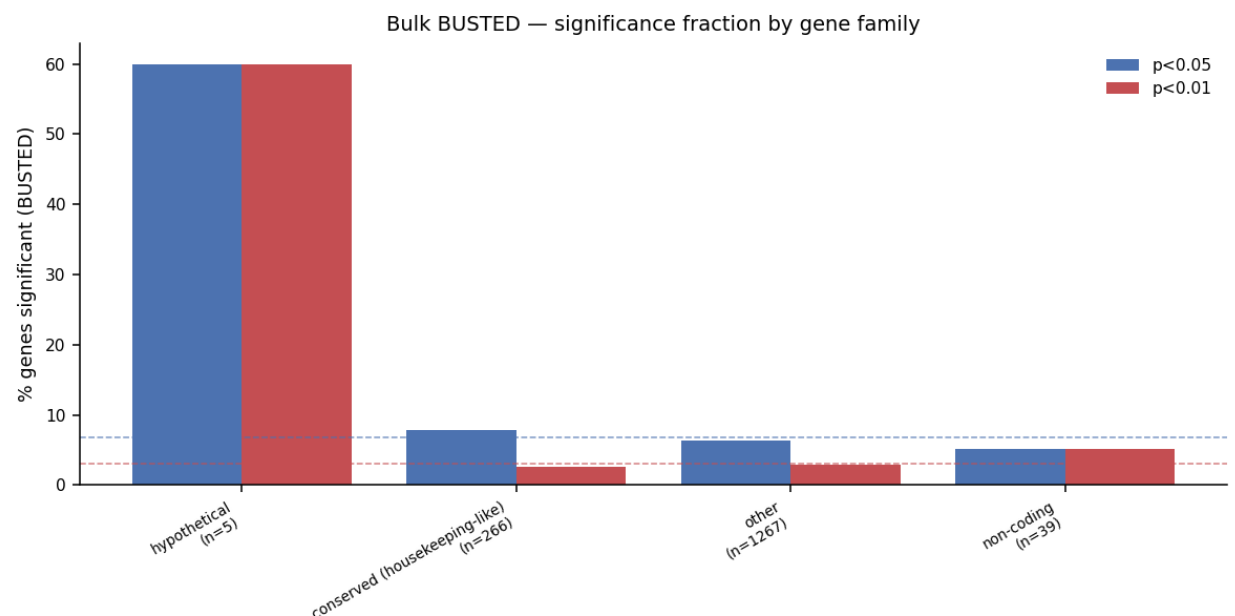


Figure 3: Family-level BUSTED enrichment

By family bucket the rates at $p<0.05$ are: hypothetical 60% ($n=5$, small denominator with large variance); conserved (housekeeping-like) 8% ($n=266$); other 6% ($n=1267$); non-coding 5% ($n=39$). The baseline across all 1,584 bulk genes is 6.8% at $p<0.05$ and 3.1% at $p<0.01$ (dashed lines on Fig. 3). Variant-antigen families (PIR, PHIST, MSP, DBP, RBP, AMA) are dramatically under-represented in this bulk set—only a handful of singletons survive the 8-way single-copy orthology filter—so the family bucket plot reflects the conserved core, not the antigen-diversifying periphery. Within the core, the “conserved” housekeeping-like bucket tracks the baseline at 7.9%, modestly elevated, in line with episodic selection on a subset of conserved enzymes and chaperones. The hypothetical-bucket spike rides on five genes only and should be re-examined once an inframe domain annotation pass is available. The variant-antigen story belongs to the priority bundle and the coverage-gap discussion (Sections 2-3).

1.4 By functional category (Fig. 4)

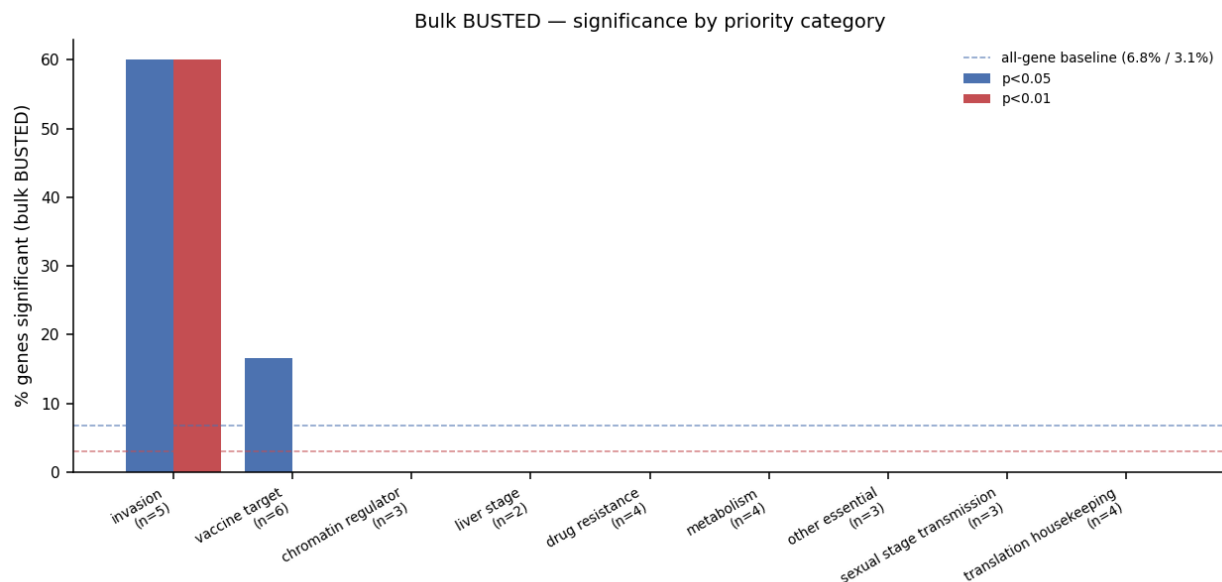


Figure 4: Bulk BUSTED enrichment by priority category

Cross-referencing the bulk p-values against our priority category table reproduces the same pattern at the functional-annotation level: invasion (60% / n=5) and vaccine target (17% / n=6) sit above the 6.8% baseline; chromatin regulator (0% / n=3), liver stage (0% / n=2), and drug resistance (0% / n=4) sit below it. Translation-housekeeping, our internal negative control, sits at the baseline floor, as it should. With per-category n in the single digits these point estimates are noisy, and we read them as direction-of-effect indicators, not effect sizes.

2. Priority bundle (41 genes)

2.1 Overview heatmap (Fig. 5)

We rank the 41 priority genes by the mean $-\log_{10}(p)$ across BUSTED, aBSREL, and MEME (panel a), with the FEL positively-selected codon count attached as a separate column (panel b). Asterisks mark cells crossing the conventional $p < 0.05$ threshold. 3 of 41 cross BUSTED, 5 of 41 cross aBSREL, 8 of 41 cross MEME, and 1 of 41 (Pvs230) carries at least one positively-selected codon under FEL.

2.2 Multi-method evidence

We score each priority gene by the number of methods at which it crosses $p < 0.05$ (with FEL counted on the presence of ≥ 1 positively-selected codon). 5 of 41 genes are flagged in ≥ 2 methods. The leaders by this score are:

Gene	PVP01 id	Category	BUSTED p	aBSREL min p	MEME min p	FEL+ sites	# meth-ods sig
Pvs230	PVP01_0415800	vaccine_target	5.59e-02	2.69e-02	1.38e-03	1	3
AARP	PVP01_0531900	invasion	1.68e-02	3.78e-06	1.81e-02	0	3
RON2	PVP01_1255000	invasion	1.12e-03	5.63e-06	1.34e-02	0	3
AMA1	PVP01_0934200	vaccine_target	3.90e-01	1.53e-03	3.60e-02	0	2
CSS	PVP01_1344100	invasion	3.92e-04	2.91e-04	1.41e-01	0	2

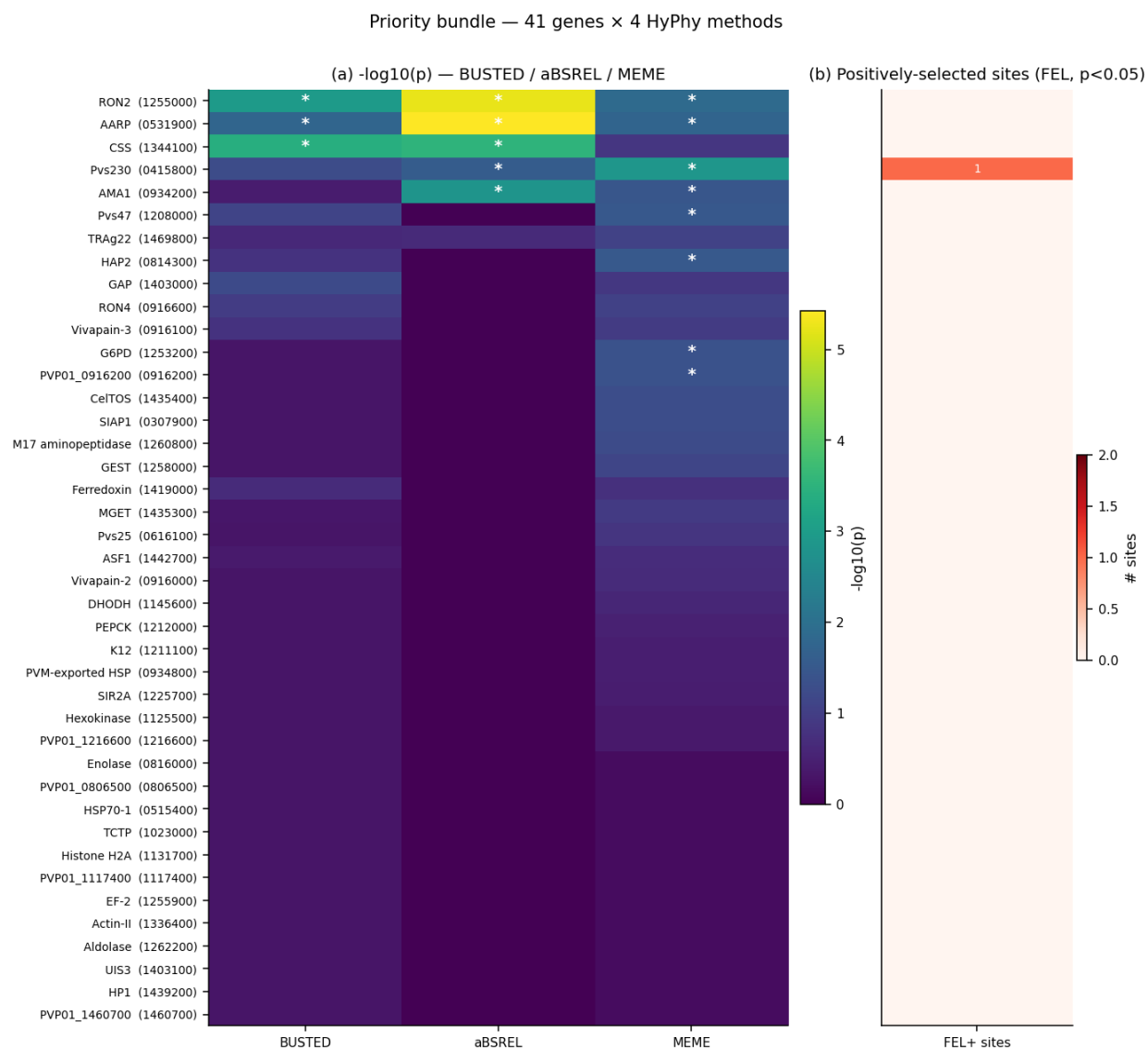


Figure 5: Priority-bundle heatmap: 41 genes × 4 HyPhy methods

These are the cleanest selection candidates we have under the present alignment-and-tree pipeline. Pvs230 crosses BUSTED at $p=0.056$ only; the call here is conservative, but aBSREL, MEME, and FEL all flag it. The Pvs230 BUSTED p in the priority bundle differs from its bulk-screen value because the priority pipeline retrims the alignment and re-fits the gene tree before HyPhy; bulk and priority p -columns are not identical alignments. The full 41-row table is in Section 6.

2.3 Vaccine targets — selection landscape (Fig. 6)

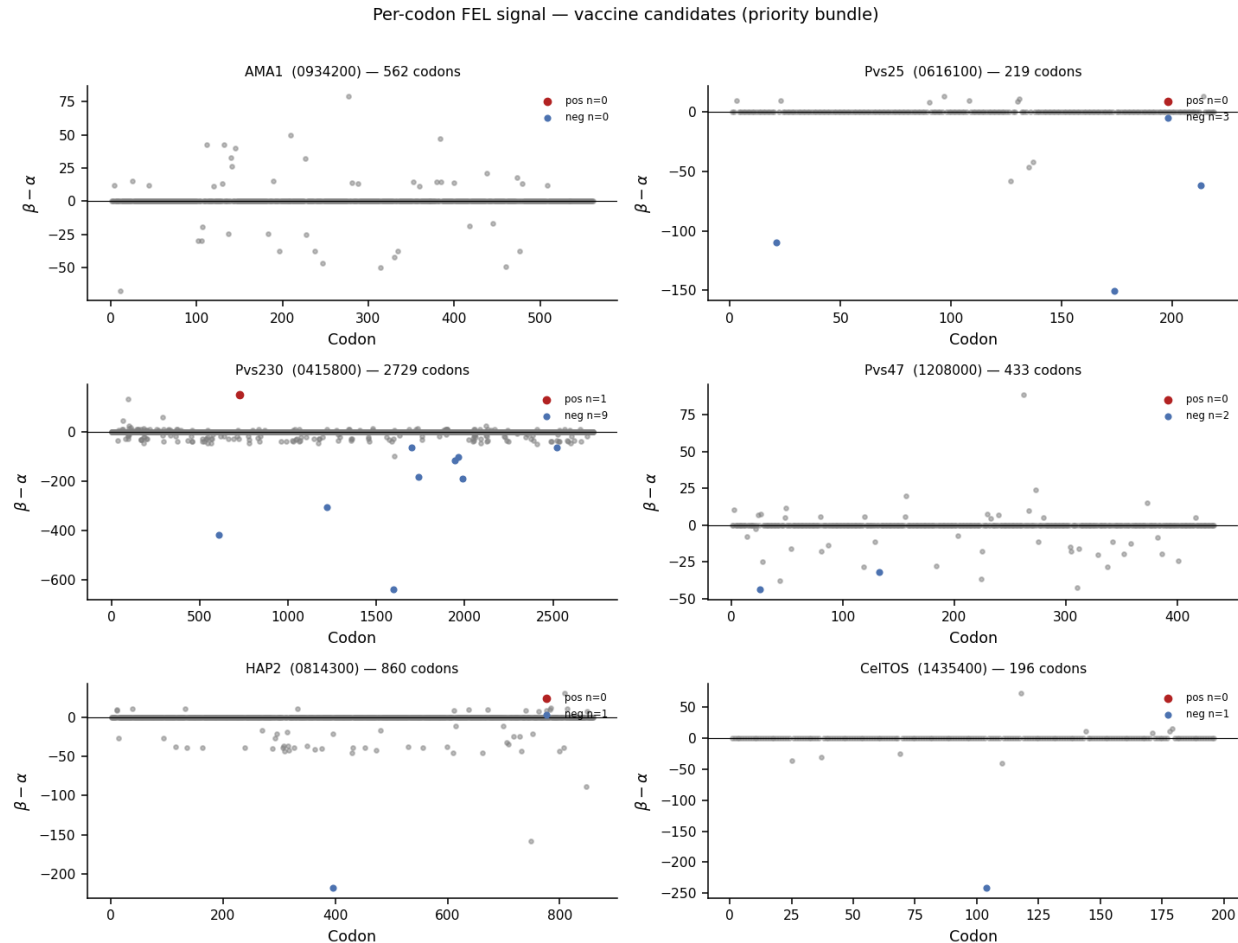


Figure 6: Per-codon FEL signal for vaccine-candidate priority genes

The vaccine-target set has 6 members with surviving HyPhy output. We plot per-codon $\beta - \alpha$ with FEL $p < 0.05$ sites coloured. Only Pvs230 carries a FEL-positive codon (codon ~720); the rest carry purifying-only sites or no FEL hits. Most vaccine candidates—CSP, MSP1, the DBP/RBP family, Pvs28, Pvs48/45—are not in this panel because their alignments did not pass Phase H; the genes shown here are the survivors and should be treated as anecdotal in the absence of the full set (Section 3).

2.4 Invasion proteins (Fig. 7)

Invasion factors—rhoptry-neck proteins, AARP, CSS, and erythrocyte-binding paralogues that survived Phase H—show scattered codons of large $\beta - \alpha$ excursion embedded in a purifying-dominated background, but the FEL $p < 0.05$ set is dominated by purifying sites; no priority invasion gene carries a FEL-positive codon. The aBSREL branch signal is strong (RON2, AARP, AMA1, CSS each have ≥ 1 significant branch) and consistent with host-receptor arms-race dynamics reported across *Plasmodium* spp.

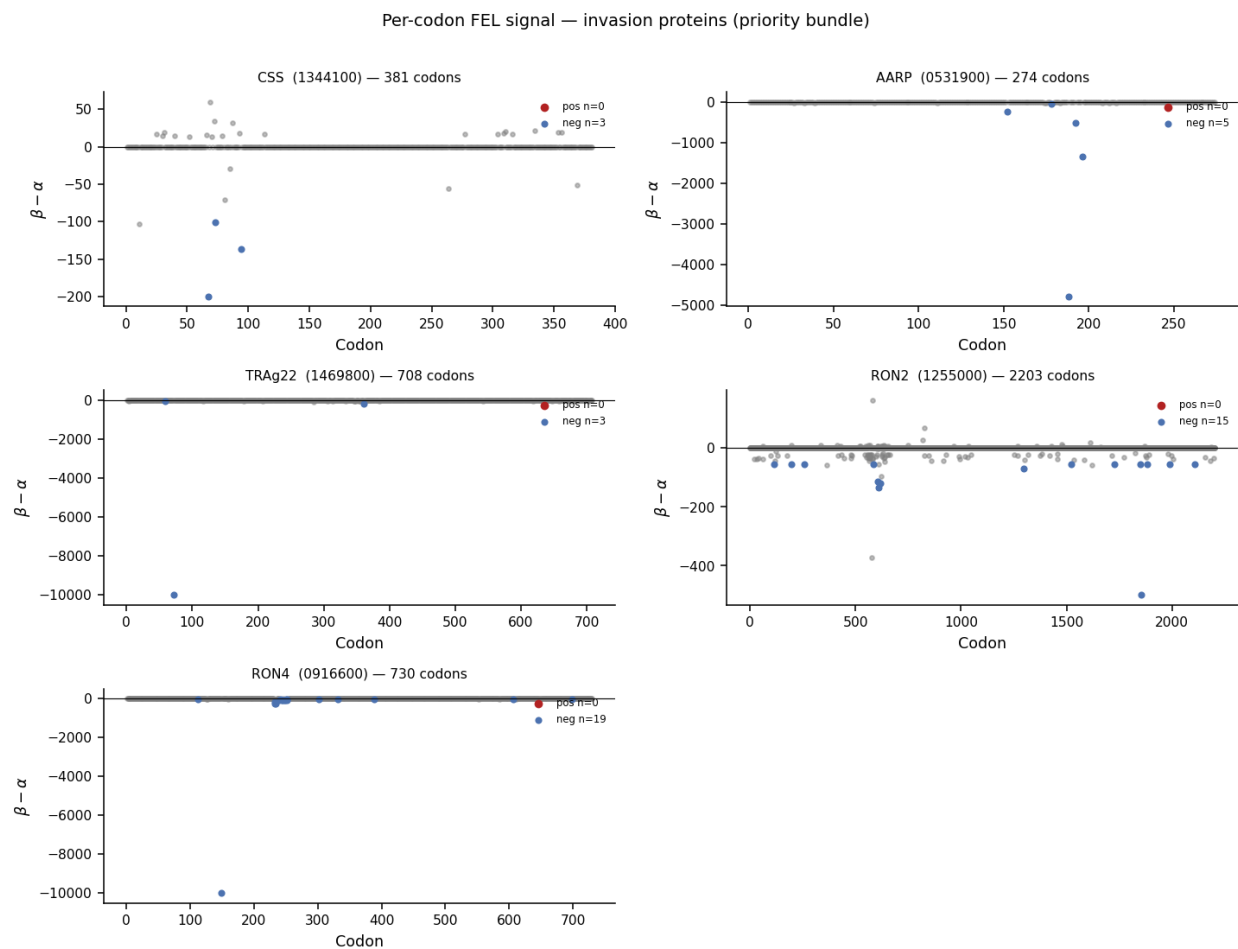


Figure 7: Per-codon FEL signal for invasion / erythrocyte-binding priority genes

2.5 Housekeeping controls (Fig. 8)

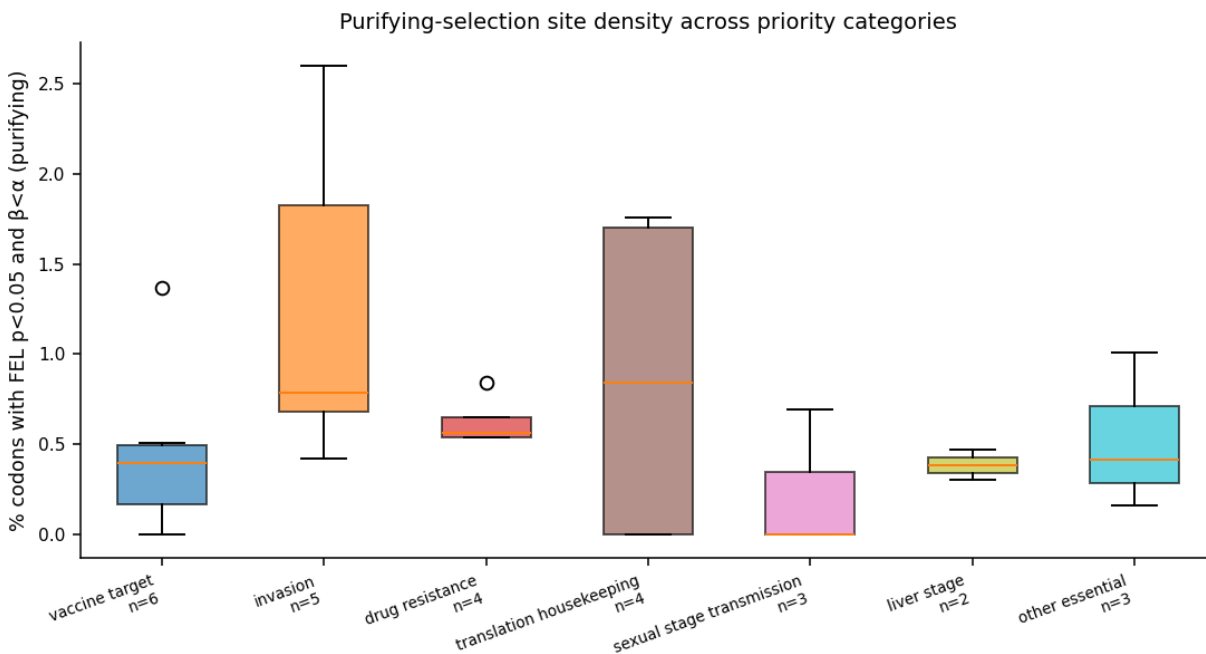


Figure 8: Purifying-site density by priority category

We use the FEL—site density—fraction of codons with $p < 0.05$ and $\beta < \alpha$ —as a proxy for purifying pressure. Invasion and translation-housekeeping categories carry the highest medians and the broadest ranges (each spans 0 to ~1.8% of codons); vaccine targets carry the lowest median (~0.4%) with a single high outlier (Pvs230). The pattern is broadly consistent with relaxed purifying constraint on antigenic surfaces and tight constraint on housekeeping. Per-category n is 2–6 and Pvs230 is large enough to drag the vaccine-target distribution alone, so we read Fig. 8 as a sanity check, not a quantitative statement (Section 4).

3. Coverage gaps

The pipeline attempted 157 priority HyPhy bundles in total; only 41 produced output, leaving 116 missing. Phase H rejects a gene when the codon alignment fails one of several quality filters—fewer than 6 strains intact, premature stop codons, frame drift, or trimAl collapse below a length floor. The 116 missing bundles fall into the following functional categories (21 of the 116 trace to candidate IDs that are not in the canonical priority table and are listed as “unmapped”):

Category	N genes missing	Example symbols
unmapped (not in priority table)	21	PVP01_1330600, PVP01_1132600, PVP01_0416800, PVP01_0918800, PVP01_1031200
drug resistance	14	DHPS, MDR1, ATP4, FNT, Plasmepsin V, Plasmepsin IX, Plasmepsin X, DHFS-FPGS
vaccine target	12	DBP, MSP3-beta, MSP3-gamma, MSP7, MSP9, MSP10, Pvs28, Pvs48/45
translation housekeeping	11	Alpha-tubulin, Beta-tubulin, Actin-I, EF-1alpha, EF-Tu, Calmodulin, Profilin, Histone H4

Category	N genes missing	Example symbols
invasion	9	MAEBL, RipR, PTRAMP, RAP1, RAP2, RhopH3, RON5, TRAg38
sexual stage transmission	8	AP2-G, AP2-G2, AP2-O, GDV1, ULG8, Pvs16, g377, PUF2
erythrocyte binding	7	EBP, RBP1a, RBP1b, RBP2a, RBP2b, RBP2c, RBP2-P1
variant antigen marker	6	PHIST-a (exemplar), PHIST-b (exemplar), STP1, VIR-C exemplar, ETRAMP11.2, Pv-fam-h (exemplar)
other essential	6	SUB1, SUB2, SERA4, SERA5, DPAP1, MSP-merozoite organizer
surface antigen	6	MSP4, MSP5, MSP8, Pv12, Pv38, Pv41
liver stage	5	SPELD, LISP2, UIS4, P36, P52
chromatin regulator	3	GCN5, SET2, BDP1
apicoplast marker	3	LytB, IspD, FabZ
cytoadherence	3	KAHRP-like, SBP1-like, VIR-cytoadherence exemplar
metabolism	2	LDH, GAPDH

The list is heavy on exactly the targets a clinical reader would care about—the vaccine candidates DBP / MSP3 / MSP7 / MSP9 / MSP10 / Pvs28 / Pvs48-45 / RBP1a-b / RBP2a-b-c / RBP2-P1, the drug-resistance markers DHPS / MDR1 / ATP4 / FNT / Plasmepsin V / IX / X / DHFS-FPGS, and the liver-stage factors P36 / P52 / UIS4 / LISP2 / SPELD. We discuss the remedy in Section 5.

4. Caveats

- **No GARD pre-screen.** Recombination breakpoints within a gene inflate dN/dS-style false positives. We do not screen for them here, and any single BUSTED/MEME hit on a paralogue-prone family (PIR, PHIST, MSP, DBP, RBP) should be interpreted with this in mind.
- **8 strains is a small tree.** aBSREL and MEME—both branch-/site-level methods—lose power on shallow trees, and our 7-tip ingroup (PvP01 is the reference; 7 query strains) sits well below the ~20-tip comfort zone of these tools.
- **No outgroup.** All HyPhy runs here are unrooted *P. vivax*-only fits. We cannot polarise lineage-specific signals against a *P. knowlesi* or *P. cynomolgi* outgroup until we add one.
- **Liftoff-annotation bias.** For PvSY56, PvT01, PvC01, and MHC087 the gene models are projected via Liftoff/TOGA2 rather than annotated *de novo*. Frame errors at the projection step are the dominant cause of the 116 dropped priority genes in Section 3.
- **Small per-category n.** Several Fig. 4 and Fig. 8 buckets are 2–6 genes. Direction of effect, not effect size, is the appropriate reading.

5. Recommendations

1. **GARD recombination pre-screen** on all 41 priority bundle alignments before publishing any positively-selected-site claim.
2. **Relaxed Phase H rebuild** at min_intact=5 (currently 7) to recover the 116 missing priority genes—the rebuild trades 2 strains of breadth for the full vaccine / drug-resistance / liver-stage panel.
3. **HyPhy RELAX** between Sal-I (lab-adapted) and the 7 field/clinical strains to test for relaxation vs intensification of selection on the lab line.
4. **McDonald-Kreitman** with the MalariaGEN *P. vivax* genomic-epidemiology VCFs—within-species polymorphism vs cross-strain divergence—for the high-priority hits that survive recommendations 1-2. Appendix B gives a first-pass realisation of this against the Pvs230 hit.

6. Tables

6.1 Top 50 bulk BUSTED hits

Rank	Gene	PVP01 id	Chr	Family	Category	p (BUSTED)	-log10(p)
1	—	PVP01_1255400	chr12	other	non-priority	0.00e+00	300.00
2	—	PVP01_0815500	chr8	other	non-priority	1.18e-09	8.93
3	—	PVP01_0929400	chr9	other	non-priority	3.04e-07	6.52
4	—	PVP01_0104700	chr1	other	non-priority	1.86e-06	5.73
5	—	PVP01_1027200	chr10	other	non-priority	3.89e-06	5.41
6	—	PVP01_1219200	chr12	hypothetical	non-priority	1.12e-05	4.95
7	—	PVP01_1110200	chr11	other	non-priority	2.89e-05	4.54
8	—	PVP01_1220100	chr12	hypothetical	non-priority	2.92e-05	4.53
9	—	PVP01_1135000	chr11	other	non-priority	3.22e-05	4.49
10	—	PVP01_0306600	chr3	other	non-priority	4.06e-05	4.39
11	—	PVP01_1258300	chr12	other	non-priority	6.89e-05	4.16
12	—	PVP01_1227100	chr12	other	non-priority	7.31e-05	4.14
13	—	PVP01_1334300	chr13	other	non-priority	1.08e-04	3.97
14	—	PVP01_1120000	chr11	other	non-priority	1.98e-04	3.70
15	—	PVP01_1323000	chr13	other	non-priority	2.26e-04	3.65
16	—	PVP01_0109200	chr1	other	non-priority	2.64e-04	3.58
17	CSS	PVP01_1344100	chr13	other	invasion	3.92e-04	3.41
18	—	PVP01_0115400	chr1	other	non-priority	4.52e-04	3.34
19	—	PVP01_0909900	chr9	conserved (housekeeping- like)	non-priority	8.14e-04	3.09
20	—	PVP01_0715700	chr7	other	non-priority	9.04e-04	3.04
21	RON2	PVP01_1255000	chr12	other	invasion	1.13e-03	2.95
22	—	PVP01_0107900	chr1	other	non-priority	1.51e-03	2.82
23	—	PVP01_1252800	chr12	conserved (housekeeping- like)	non-priority	2.06e-03	2.69
24	—	PVP01_0616400	chr6	other	non-priority	2.35e-03	2.63
25	—	PVP01_0409900	chr4	other	non-priority	2.85e-03	2.54
26	—	PVP01_1427700	chr14	other	non-priority	3.01e-03	2.52
27	AARP	PVP01_0531900	chr5	other	invasion	3.13e-03	2.50
28	—	PVP01_1464900	chr14	other	non-priority	3.16e-03	2.50
29	tRNA	PVP01_0815300	chr8	non- coding	non-priority	3.30e-03	2.48
30	—	PVP01_0913000	chr9	other	non-priority	3.70e-03	2.43
31	—	PVP01_0926500	chr9	other	non-priority	3.83e-03	2.42
32	—	PVP01_0944000	chr9	other	non-priority	4.24e-03	2.37
33	—	PVP01_1406800	chr14	conserved (housekeeping- like)	non-priority	4.66e-03	2.33
34	—	PVP01_1341500	chr13	conserved (housekeeping- like)	non-priority	4.71e-03	2.33
35	—	PVP01_0707500	chr7	conserved (housekeeping- like)	non-priority	4.89e-03	2.31
36	—	PVP01_1455000	chr14	conserved (housekeeping- like)	non-priority	5.77e-03	2.24

Rank	Gene	PVP01 id	Chr	Family	Category	p (BUSTED)	-log10(p)
37	—	PVP01_1107700	chr11	other	non-priority	5.81e-03	2.24
38	—	PVP01_1243600	chr12	other	non-priority	5.85e-03	2.23
39	—	PVP01_0815600	chr8	other	non-priority	5.96e-03	2.22
40	—	PVP01_1469500	chr14	hypothetical	non-priority	6.21e-03	2.21
41	—	PVP01_1444900	chr14	other	non-priority	6.90e-03	2.16
42	—	PVP01_1133300	chr11	other	non-priority	6.92e-03	2.16
43	—	PVP01_1267500	chr12	other	non-priority	7.05e-03	2.15
44	—	PVP01_0214500	chr2	conserved (housekeeping-like)	non-priority	7.53e-03	2.12
45	—	PVP01_0213000	chr2	other	non-priority	9.19e-03	2.04
46	—	PVP01_1320300	chr13	other	non-priority	9.30e-03	2.03
47	—	PVP01_1121800	chr11	other	non-priority	9.37e-03	2.03
48	—	PVP01_1142800	chr11	other	non-priority	9.93e-03	2.00
49	rRNA	PVP01_1117500	chr11	non-coding	non-priority	9.94e-03	2.00
50	—	PVP01_1262300	chr12	other	non-priority	1.03e-02	1.99

(Full top-200 table at [writeup/top200_bulk_busted.tsv](#).)

6.2 Priority bundle — all 41 genes, four methods

Gene	PVP01 id	Category	BUSTED p	aBSREL min p	MEME min p	FEL+	FEL−	n sites	n branches	sig (aB- SREL)
SIAP1	PVP01_0307900	chr_stage	5.00e-01	1.00e+00	5.25e-02	0	3	997	0/10	
Pvs230	PVP01_0415800	chr_cine_target	5.59e-02	2.69e-02	1.38e-03	1	9	2729	1/13	
HSP70-1	PVP01_0515400	translation_housekeeping	5.00e-01	1.00e+00	6.67e-01	0	12	682	0/8	
AARP	PVP01_0531900	chr_cine	1.68e-02	3.78e-06	1.81e-02	0	5	274	1/12	
Pvs25	PVP01_0616400	chr_cine_target	5.00e-01	1.00e+00	1.44e-01	0	3	219	0/12	
PVP01_0803600	PVP01_0803600	chr_cine	5.00e-01	1.00e+00	6.67e-01	0	1	250	0/8	
HAP2	PVP01_0814800	chr_cine_target	1.60e-01	1.00e+00	3.13e-02	0	1	860	0/10	
Enolase	PVP01_0816000	chr_cine	5.00e-01	1.00e+00	6.67e-01	0	3	446	0/9	
Vivapain2	PVP01_0916600	chr_cine_essential	5.00e-01	1.00e+00	2.15e-01	0	2	484	0/8	
Vivapain3	PVP01_0916400	chr_cine_essential	1.64e-01	1.00e+00	1.15e-01	0	5	495	0/11	
PVP01_0911300	PVP01_0911300	chr_cine	5.00e-01	1.00e+00	4.37e-02	0	6	487	0/9	
RON4	PVP01_0916300	chr_cine	1.06e-01	1.00e+00	8.91e-02	0	19	730	0/11	
AMA1	PVP01_0934200	chr_cine_target	3.90e-01	1.53e-03	3.60e-02	0	0	562	2/13	
PVM-exported	PVP01_0934800	chr_cine	5.00e-01	1.00e+00	3.57e-01	0	2	663	0/8	
HSP										
TCTP	PVP01_1023000	chr_cine_resistance	5.00e-01	1.00e+00	6.67e-01	0	1	171	0/9	
PVP01_1117400	PVP01_1117400	chr_cine	5.00e-01	1.00e+00	6.67e-01	0	0	181	0/8	
Hexokinase	PVP01_1125500	chr_cine	5.00e-01	1.00e+00	4.19e-01	0	4	493	0/8	
Histone H2A	PVP01_1131700	translation_housekeeping	5.00e-01	1.00e+00	6.67e-01	0	0	133	0/8	

Gene	PVP01 id	Category	BUSTED p	aBSREL min p	MEME min p	FEL+	FEL−	n sites	n branches sig (aB- SREL)
DHODHP	PVP01_1145600	g_resistance	5.00e-01	1.00e+00	2.57e-01	0	3	557	0/10
Pvs47	PVP01_1208000	cine_target	7.59e-02	1.00e+00	3.15e-02	0	2	433	0/13
K12	PVP01_1211400	g_resistance	5.00e-01	1.00e+00	3.51e-01	0	6	712	0/8
PEPCK	PVP01_1212000	abolism	5.00e-01	1.00e+00	3.25e-01	0	2	599	0/8
PVP01_P118600	PVP01_1216600	er	5.00e-01	1.00e+00	4.28e-01	0	6	513	0/10
SIR2A	PVP01_1225700	matin_regul	5.00e-01	1.00e+00	3.71e-01	0	2	306	0/8
G6PD	PVP01_1253200	g_resistance	5.00e-01	1.00e+00	4.33e-02	0	5	927	0/9
RON2	PVP01_1255000	asion	1.12e-03	5.63e-06	1.34e-02	0	15	2203	1/12
EF-2	PVP01_1255000	slation_housekeepi	5.00e-01	1.00e+00	6.67e-01	0	14	832	0/9
GEST	PVP01_1258000	ual_stage_transmission	5.00e-01	1.00e+00	7.50e-02	0	0	249	0/9
M17	PVP01_1260800	er_essential	5.00e-01	1.00e+00	5.69e-02	0	1	621	0/8
aminopep- ti- dase									
Aldolase	PVP01_1262200	abolism	5.00e-01	1.00e+00	6.67e-01	0	5	369	0/9
Actin- II	PVP01_1336400	slation_housekeepi	5.00e-01	1.00e+00	6.67e-01	0	0	376	0/8
CSS	PVP01_1344100	asion	3.92e-04	2.91e-04	1.41e-01	0	3	381	1/12
GAP	PVP01_1403000	ual_stage_transmission	5.00e-01	1.00e+00	1.32e-01	0	0	244	0/11
UIS3	PVP01_1403100	er_stage	5.00e-01	1.00e+00	6.67e-01	0	1	213	0/8
Ferredoxin	PVP01_1419000	oplast_mark	2.19e-01	1.00e+00	1.80e-01	0	1	196	0/8
MGET	PVP01_1435800	ual_stage_transmission	5.00e-01	1.00e+00	1.20e-01	0	2	288	0/9
CelTOS	PVP01_1435400	cine_target	5.00e-01	1.00e+00	5.17e-02	0	1	196	0/9
HP1	PVP01_1439200	matin_regul	5.00e-01	1.00e+00	6.67e-01	0	0	275	0/8
ASF1	PVP01_1442700	matin_regul	3.99e-01	1.00e+00	2.02e-01	0	2	269	0/9
PVP01_P143700	PVP01_1460700	er	5.00e-01	1.00e+00	6.67e-01	0	0	115	0/8
TRAg22	PVP01_1469800	asion	2.39e-01	2.29e-01	8.61e-02	0	3	708	0/11

(Full priority bundle table at `writeup/priority_bundle_results.tsv`.)

Appendix A. Pvs230 per-codon hits mapped to ICP / IVP / C-terminal domains

Pvs230—our top multi-method priority hit—warrants a closer look against the published clinical-isolate diversity literature [8, 9]. Doi et al. 2011 surveyed 113 *pvs230* sequences worldwide and reported low nucleotide diversity overall ($\theta\pi = 0.00118$) with diversity concentrated in a short N-terminal interspecies variable part (IVP). Feng et al. 2022 refined this picture on China-Myanmar border and central Myanmar isolates: codon-based tests recovered positive selection on the IVP (codons 1-269) and purifying selection on the ICP (codons 270-954). We mapped our 8-strain HyPhy hits onto the same coordinate system using the PVP01 reference protein (2,725 residues; CDS LT635615.1:636,087-644,264, minus strand, single exon).

MEME (episodic positive selection, $p < 0.05$). We recover 6 sites; mapping to PVP01 residues: 65, 102, 112, 252, 720, and 2,080. Four of the six (residues 65, 102, 112, 252) fall inside the IVP (codons 1-269)—the same domain Feng et al. flagged as positively selected on hundreds of clinical sequences—reproducing the published clinical signal from an 8-strain reference panel alone. Residue 720 lies in the ICP, and residue 2,080 lies in the C-terminal region beyond the segment surveyed by either Doi or Feng (both papers truncate at ~codon 954).

FEL (per-site, pervasive selection). We recover 1 positively-selected codon (residue 720, $p = 0.042$) and 9 purifying codons—residues 604, 1,215, 1,595, 1,697, 1,735, 1,940, 1,960, 1,983, and 2,520. Every

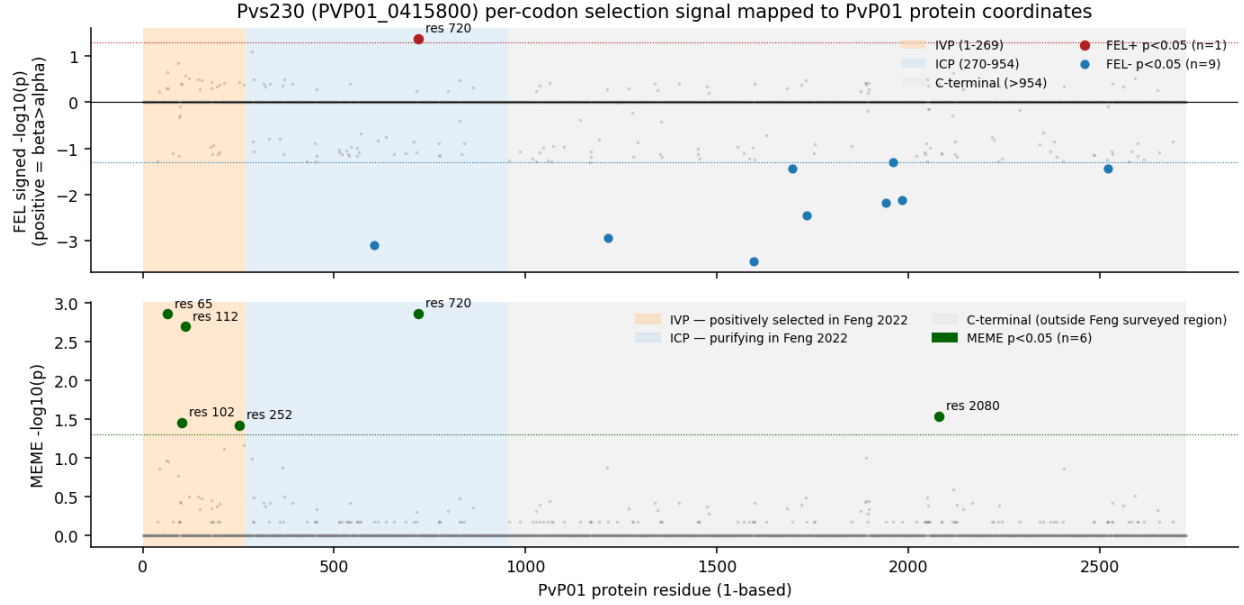


Figure 9: Pvs230 per-codon FEL and MEME signal across PvP01 protein coordinates with IVP/ICP/C-terminal shading

purifying FEL hit lies in the ICP or the C-terminal extension—consistent with Feng’s purifying-selection ICP call—and the single FEL+ codon (720) overlaps the ICP MEME signal, indicating that residue 720 carries an episodic-and-pervasive double signature rather than a transient lineage-specific event.

Interpretation. Our 8-strain panel reproduces the IVP-localised diversifying signal and the ICP-localised purifying signal at the residue-level resolution of the published clinical-isolate work, and it adds a residue (2,080) in the C-terminal region that prior work did not survey. The 2,080 site is a natural candidate for follow-up at full clinical-cohort depth—it is far enough downstream of the canonical Pvs230 N-terminal vaccine fragment that any selection signature there has direct implications for whole-protein TBV constructs. We do not over-interpret a single 8-strain MEME hit, and we flag the codon as a follow-up target rather than a finding.

Appendix B. Pvs230 SNP landscape from MalariaGEN Pv4 (1,895 samples)

We extracted the Pvs230 locus (LT635615.1:636,087-644,264, single exon, minus strand) from the MalariaGEN Pv4 chr04 callset (1,895 samples, GATK ApplyRecalibration release), kept PASS-filter biallelic SNPs, and classified each per-alt allele as synonymous, non-synonymous, or stop-gain against the PvP01 reading frame. 306 SNP records survived: 128 synonymous, 176 non-synonymous, and 2 stop-gain.

Domain-stratified N/S is striking. The IVP (1-269), ICP (270-954), and C-terminal extension (>954) yield N/S ratios of 6.18 / 1.04 / 0.89 and SNP densities of 29.7 / 8.0 / 9.7 SNPs per 100 aa. The IVP carries 68 non-synonymous SNPs against 11 synonymous—a 6:1 excess of replacement variation, the textbook signature of diversifying selection—while the ICP and C-terminal run near or below the syn=non-syn line, consistent with purifying selection. This recovers Doi 2011’s worldwide IVP-vs-ICP picture and Feng 2022’s codon-level positive/purifying split at clinical-cohort depth from a single VCF rather than 113 or 92 manually-curated sequences.

Direct MalariaGEN support for the 8-strain HyPhy hits. Mapping the 1,895-sample MalariaGEN SNPs onto our 6 MEME / 1 FEL+ / 9 FEL- residues:

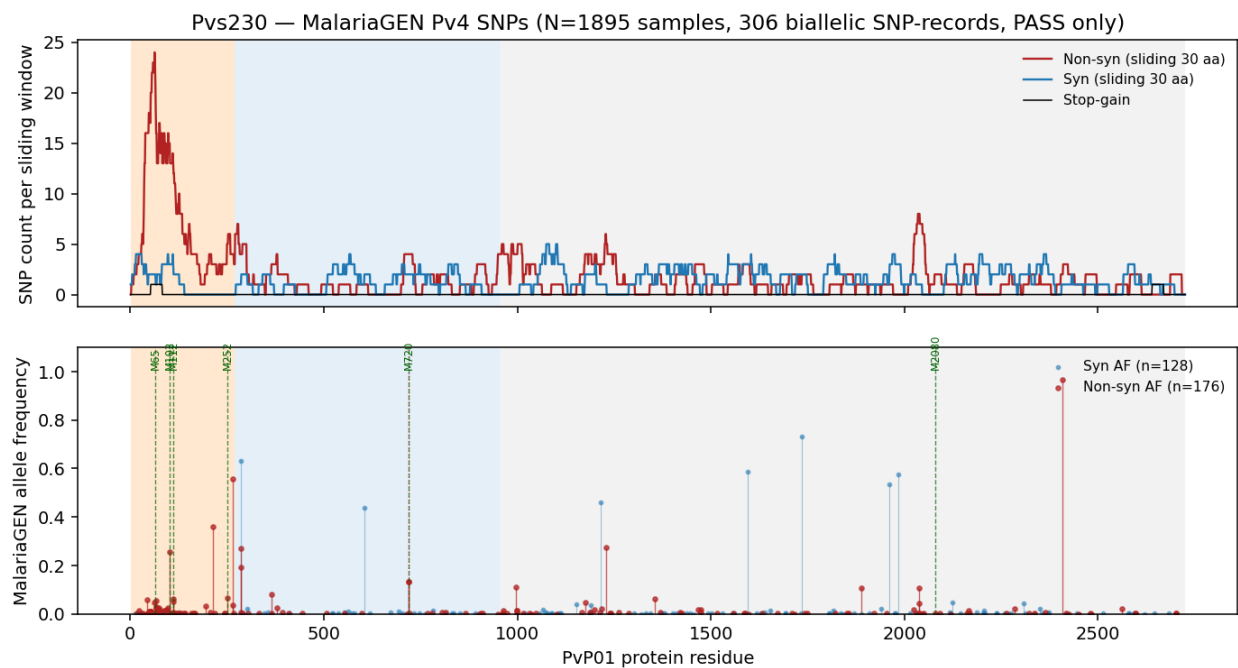


Figure 10: Pvs230 SNP distribution along the protein — sliding 30-aa window of synonymous, non-synonymous, and stop-gain counts (top); per-SNP allele-frequency lollipops with MEME-hit residues marked (bottom)

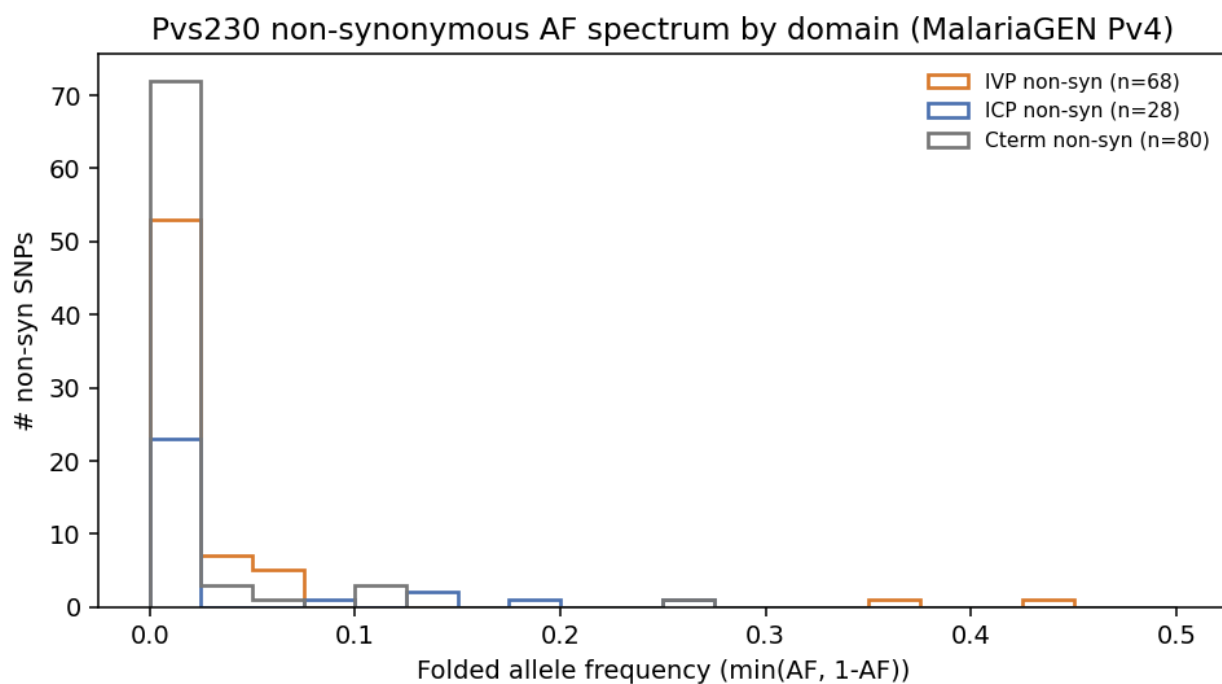


Figure 11: Folded allele-frequency spectrum of non-synonymous Pvs230 SNPs by domain

HyPhy hit	PvP01 residue	Domain	MalariaGEN SNP(s) at codon	Field AF	Reading
MEME	65	IVP	—	—	No field SNP at site or ± 1 flank — selection signal sits on a residue that is invariant in MalariaGEN Pv4.
MEME	102	IVP	C102Y, C102R	25.5% , 0.3%	High-frequency cysteine-to-tyrosine substitution; affects the IVP cysteine-rich region.
MEME	112	IVP	R112H, R112C	6.3%, 5.1%	Two independent non-synonymous alleles at the same codon — replicate non-synonymous events.
MEME	252	IVP	V252M	6.5%	Common non-synonymous at the IVP/ICP boundary.

HyPhy hit	PvP01 residue	Domain	MalariaGEN SNP(s) at codon	Field AF	Reading
MEME + FEL+	720	ICP	D720A + D720N (linked, encodes D720T)	13.3% (haplotype)	Adjacent- position SNPs that always co-occur (234/1,895 samples carry BOTH, zero carry one alone). The protein- level change is a single D→T sub- stitution requiring two adjacent nucleotide substitu- tions — a classic multi- nucleotide substitu- tion (MNS). See MNS callout below.
MEME	2,080	C-term	L2080I	0.3%	Rare non- synonymous; outside Doi/Feng surveyed region.

HyPhy hit	PvP01 residue	Domain	MalariaGEN SNP(s) at codon	Field AF	Reading
FEL-	604, 1215, 1595, 1735, 1940, 1960, 1983	ICP / C-term	Y604Y, G1215G, D1595D, L1735L, P1940P, A1960A, Y1983Y	2-73%	Every FEL- codon with a Malaria- GEN SNP carries a synony- mous variant only. Malaria- GEN sees nucleotide variation at these sites but only at silent positions— exactly the pattern purifying selection produces.
FEL-	1697, 2520	C-term	—	—	No field SNP.

D720 is a multi-nucleotide substitution, not two parallel single-nucleotide events. Sample-level genotype crosstab from the MalariaGEN Pv4 callset shows that the D720A allele (chr4:642106 T→G) and the D720N allele (chr4:642107 C→T) are perfectly linked—234 of 1,895 samples carry **both**, zero carry one alone, 1,366 carry neither (295 missing). In mRNA frame the reference codon GAT (Asp) becomes ACT (Thr) when both substitutions are present, so the actual protein-level change in the 234 carriers is **D720T**, a single Asp→Thr substitution that required two adjacent nucleotide substitutions—a classic multi-nucleotide substitution (MNS).

Re-running BUSTED on Pvs230 with the multi-hit option (`--multiple-hits Double`, which permits 2-nt substitutions to occur in a single evolutionary event) collapses the whole-gene signal entirely: p shifts from 0.056 (single-hit BUSTED) to 0.500 (BUSTED-MH; LRT = 0.00). The original Pvs230 whole-gene BUSTED signal was driven by this single MNS event at codon 720, and the model correctly identifies it as a single substitution once allowed to. The per-codon D→T change is still a real diversifying event present in 12.3% of field samples (~234 / ~1,600 callable), but the report should describe it as **one substitution event at moderate frequency**, not two parallel diversifying alleles. The IVP per-site MEME hits (residues 65, 102, 112, 252) are at independent codons with independent single-nucleotide SNPs and remain unaffected.

The FEL– / synonymous-only pattern is the cleanest validation in the appendix. Seven of the nine FEL– residues carry a MalariaGEN SNP, and in every single case the SNP is synonymous—frequencies up to 73%. MalariaGEN actively rejects every non-synonymous substitution at these sites while tolerating arbitrary synonymous diversification, which is what a per-codon purifying-selection call is supposed to mean. Field data and 8-strain HyPhy converge on the same residues with no parameter tuning.

Caveats. The MalariaGEN AF analysis treats each per-alt allele independently and does not correct for the

Pv4 sample geographic stratification (Asia ~70%, Africa ~5%, Americas ~25%). A formal McDonald-Kreitman test against the 8-strain divergence axis is the next step. The MNS finding at codon 720 generalises—several of our other top BUSTED hits should be re-run with `--multiple-hits Double` before publishing whole-gene p-values, since adjacent-position SNPs are not rare in *P. vivax*.

Underlying numbers in `writeup/pvs230_snp_summary.json` and per-SNP table in `writeup/pvs230_snp_table.tsv`.

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